Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering and Mathematics

School of Mathematics

JS and SS Mathematics **Semester 2 2023**
SS Theoretical Physics

MAU44400 — Quantum Field Theory

29/04/2023 RDS-SIM COURT 14:00–17:00

Professor Samson L. Shatashvili

Instructions to Candidates:

Credit will be given for the best two questions from Part A (Questions 1, 2 and 3) and the best two questions from Part B (Questions 4, 5 and 6).

All questions have equal weight.

Materials Permitted for this Examination:

Formulae and Tables tables are available from the invigilators, if required. Non-programmable calculators are permitted for this examination,—please indicate the make and model of your calculator on each answer book used.

You may not start this examination until you are instructed to do so by the Invigilator.

Part A 1. Massive gauge field

a) 10 Marks: Derive the equations of motion for the following Action functional:

$$A(\phi^\mu, \partial_\nu \phi^\mu) = \int d^4x \left[-\frac{1}{2}(\partial_\mu \phi^\nu)^2 + \frac{1}{2}(\partial_\mu \phi^\mu)^2 + \frac{m^2}{2}\phi_\mu \phi^\mu \right] =$$
$$= \int d^4x \left[-\frac{1}{4}(\partial_\mu \phi_\nu - \partial_\nu \phi_\mu)(\partial^\mu \phi^\nu - \partial^\nu \phi^\mu) + \frac{m^2}{2}\phi_\mu \phi^\mu \right]$$

b) 5 Marks: Show that these equations imply $\partial_\mu \phi^\mu = 0$.

c) 10 Marks: Construct the Hamiltonian density, and constraints if there are such, as functionals of $\phi_\mu, \partial_i \phi_\mu$ and generalized momentum Π_μ.

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2. $SO(N)$ invariant scalar field theory: 25 marks

Consider an $SO(N)$ symmetric theory of N real scalar fields with Lagrangian:

$$L = \frac{1}{2}\sum_{a=1}^{N} \partial_\mu \Phi^a \partial^\mu \Phi^a - \frac{m^2}{2}\sum_{a=1}^{N} \Phi^a \Phi^a - \frac{\lambda}{4}(\sum_{a=1}^{N} \Phi^a \Phi^a)^2$$

a) 5 marks: Derive the classical field equations

b) 5 marks: Introduce conjugate momenta $\Pi^a(x)$ (satisfying usual Poisson brackets relations $\{\Pi^a(x), \Phi^b(y)\} = \delta^{ab}\delta^{(3)}(x-y)$) and derive the Hamiltonian density $H(\Pi, \Phi)$ as a functional of Π^a and Φ^a.

c) 5 marks: Write down the Noether currents J^μ_{ab} of the $SO(N)$ symmetry: $\Phi^a \to \Lambda^{ab}\Phi^b$, or infinitesimally $\Phi^a \to \Phi^a + \epsilon^{ab}\Phi^b$, with $\Lambda^T\Lambda = I, \quad \Lambda = I + \epsilon, \quad \epsilon^{ab} = -\epsilon^{ba}$.

d) 5 marks: Using the classical equations of motion, derived in a), show that these currents are conserved: $\partial_\mu J^\mu_{ab} = 0$

e) 5 marks: Write $SO(N)$ Noether currents J^μ_{ab} in terms of Π^a and Φ^a; check that Noether charges $Q_{ab} = \int d^3x J^0_{ab}(x)$ generate above $SO(N)$ transformations on fields under Poisson Bracket - this means that you need to show that:

$$\delta\Phi^a = \{\sum_{b,c} \epsilon^{bc} Q_{bc}, \Phi^a(x)\} = \epsilon^{ab}\Phi^b$$

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3. Supersymmetry: 25 marks

Consider the Action (χ is a left-handed, Weyl, fermion; ϕ is a complex scalar field; F is a complex "auxiliary" scalar field):

$$A = \int d^4x \mathcal{L}(\phi, \chi, F) = \int d^4x \left[\partial_\mu \phi^* \partial^\mu \phi + i\chi^+ \bar{\sigma} \cdot \partial \chi + F^* F\right]$$

a) 10 marks: Show that it is invariant under following infinitesimal transformations up to a total derivative

$$\delta\phi = 0$$

$$\delta\chi = \epsilon F$$

$$\delta F = -i\epsilon^+ \bar{\sigma} \cdot \partial \chi$$

Calculate only in the first order for ϵ, drop second order terms in ϵ during calculation. Here ϵ is anti-commuting variable (anti-commutes with itself and with other fermions, like χ). $\sigma^\mu = (1, \sigma^a)$; $\quad \bar{\sigma}^\mu = (1, -\sigma^a)$ and $\sigma \cdot \partial = \sigma^\mu \partial_\mu, \quad \bar{\sigma} \cdot \partial = \bar{\sigma}^\mu \partial_\mu$.

b) 15 marks: Show that it is also invariant up to total derivatives under the more complicated infinitesimal transformation:

$$\delta\phi = -i\epsilon^T \sigma^2 \chi$$

$$\delta\chi = \epsilon F - \sigma \cdot \partial \phi \sigma^2 \epsilon^*$$

$$\delta F = -i\epsilon^+ \bar{\sigma} \cdot \partial \chi$$

Calculate only in the first order for ϵ, drop second order terms in ϵ during calculation. Such symmetry is called **supersymmetry**.

Note 1: use following identities about Weyl fermions and Pauli matrices:

$$\sigma^2 \sigma^a \sigma^2 = -(\sigma^a)^*; \qquad a = 1, 2, 3$$

$$\sigma^2 \sigma^\mu \sigma^2 = (\bar{\sigma}^\mu)^*$$

$$\chi^T \epsilon = \chi^a \epsilon^a = -\epsilon^a \chi^a = -\epsilon^T \chi$$

$$\chi^T \sigma^2 \epsilon = \chi^a \sigma^2_{ab} \epsilon^b = -\chi^a \sigma^2_{ba} \epsilon^b = \epsilon^b \sigma^2_{ba} \chi^a = \epsilon^T \sigma^2 \chi$$

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Note 2: freely integrate by parts.

Note 3: Assuming that it is invariant under transformations from question a) the question b) is reduced to checking the symmetry only under

$$\delta\phi = -i\epsilon^T\sigma^2\chi$$

$$\delta\chi = -\sigma\cdot\partial\phi\sigma^2\epsilon^*$$

$$\delta F = 0$$

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Part B 4. Four-point functions in $\frac{\lambda}{4!}\phi^4$ theory: 25 marks

Draw all diagrams with four external legs in ϕ^4 theory up to one loop (there is one diagram in zero-loop and three diagrams in one-loop):

$$i\mathcal{M}(p_1 p_2 \to p_3 p_4) = -i\lambda + (-i\lambda)^2[iV(s) + iV(t) + iV(u)]$$

a) 10 marks: Write the expression for $V(s)$ as an integral over Feynman parameter x and loop four-momentum.

b) 15 marks: Use dimensional regularization to evaluate momentum integral (you don't have to integrate over Feynman parameter). $s = (p_1 + p_2)^2$.

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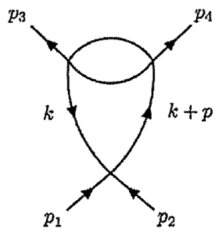
5. 2-loop in ϕ^4 theory: 25 Marks

a) 10 Marks: Write down the expression for the following 2-loop Feynman diagram $V_{[2-loop]}$ in $\frac{1}{4}\lambda\phi^4$ theory



in terms of the integral over undetermined loop momentum k and 1-loop 4-point sub-diagram $V_{[1-loop]}$ (with outgoing momenta $[p_3, p_4]$ and incoming momenta $[-k, k+p]$).

b) 5 Marks: Express $V_{[1-loop]}$ in terms of the integral over its own loop momentum.

c) 5 Marks: Use Feynman parameter representation, separately for $V_{[1-loop]}$ and $V_{[2-loop]}$ (which has been already written in a) in terms of $V_{[1-loop]}$); do not have to calculate the integrals.

d) 5 Marks: Is this 2-loop diagram divergent or convergent in four-dimensions? Explain the answer.

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6. 2d QED: 25 Marks

a) 10 marks: Write down the expression for one loop photon two-point function in d-dimensional QED (Feynman rules are the same as in 4 dimensions, but all momentum integrals are d-dimensional). Introduce the Feynman parameter and simplify the expression.

b) 10 marks: Evaluate the momentum integral after Euclidean rotation, reduce the expression to the integral over the Feynman parameter only (integral over Feynman parameter is not necessary to compute) and express it through Gamma function.

b) 5 marks: Now choose $d = 2$. Is the final answer for $d = 2$ convergent, or divergent? Explain this from the final expression.

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